Ashish Yadav

Full-Stack Engineer · Go, React & Distributed Systems

+91 875-008-4388 | ashish.devv2026@gmail.com | Portfolio | LinkedIn | GitHub | LeetCode | Codolio

PROFESSIONAL SUMMARY

Full-stack engineer building Go backends and React/Next.js frontends across FinTech, SaaS, and AI products. Shipped a 50+ domain serverless platform and a 100K+ user AI chat app, with measurable gains in latency, cost, and reliability on AWS and GCP.

EDUCATION

K.R. Mangalam University — B.Tech in Computer Science and Engineering *Aug 2023 – Jun 2027 (Expected)*

Coursework: Data Structures, Algorithms, OOD, DBMS, Operating Systems, Computer Networks, Complexity Analysis

EXPERIENCE

Software Developer Intern, ASOasis Tech *Aug 2025 – Present · Remote*

- Architected a serverless backend (Go, AWS Lambda) spanning 50+ product domains on single-table-per-domain DynamoDB, enforcing a zero-Scan read policy and 15+ shared rate limiters, in close coordination with product and engineering teams.
- Shipped a real-time AI chat backend for a 100K+ download consumer app at sub-200ms latency, pairing a Gorse recommendation engine (Redis + MySQL) with a sigmoid-based match model across 5 engagement signals.
- Engineered human-in-the-loop approval for AI agent actions on AWS Step Functions (**.waitForTaskToken**), pausing sensitive steps at zero compute cost and running side-effects only after approval, with a **transactional outbox** (single Postgres transaction + EventBridge relay) and optimistic locking keeping state consistent.
- Developed a fault-tolerant VS Code AI diff pipeline with four-tier fallback matching (exact, normalized, fuzzy-anchor, line-number) that lifted patch success from 40% to 100% while cutting inference cost ~80%.
- Built 6+ grounded-generation AI features (adaptive mock interviews, resume tailoring, promotion packets) that require every AI claim to cite a server-verified source, with guards blocking hallucinated output before it's persisted.
- Constructed a deterministic fault-injection framework with a 14-spec invariant checker over append-only event logs; it detected 15/15 seeded distributed-system bugs (7 held out as hidden) with 0 false violations across 45,102 checks.
- Implemented a staged rollout control plane in Go (Internal → 1% → 5% → 100%) on Firebase Remote Config that auto-rolls back when telemetry degrades (e.g. purchase success rate < 95%), streaming live status over SSE.
- Delivered a multi-tenant workflow automation platform (poll-based EventBridge scheduler, 2–8 concurrent workers) and a BigQuery analytics backend, cutting repeat query volume ~95% via a 5-minute TTL cache.
- Hardened production security with SSRF protection, a no-egress sandbox, PII-minimized logging, and GDPR export/erasure — traced and fixed a live rate-limit-bypass incident via a signed-header trust scheme.

Freelance Software Developer · Independent Contractor *Nov 2024 – Jul 2025 · Remote*

- Shipped Swipe It, an AI recruitment platform using vector embeddings that improved candidate-role match precision 80% over keyword search, deployed end-to-end for a recruitment startup.
- Developed an ERP backend with an admin-configurable Formula Engine reaching 100% stock-reconciliation accuracy and automating order-to-production workflows via email parsing, cutting operational overhead 90%.

PROJECTS

Mini-Dropbox — Go · REST API · Distributed Systems · GitHub *Nov 2024*

- Engineered a distributed file-storage system in Go with master-node coordination, content-addressed storage (MD5), and a REST API for registration, deduplication, and cross-node retrieval.
- Added sync.Pool buffer pooling and an in-memory existence cache, lifting concurrent throughput 4x (500→2,000 req/s) while cutting memory 43% and retrieval time 55%.
- Designed a RESTful HTTP API for file registration and cross-node retrieval, with RWMutex-protected operations ensuring safe concurrent access across both storage nodes under load.

ASHOS — AI-Powered CLI Personal OS — Go, OpenAI, RAG, SQLite, Vector Search · GitHub *Sep 2024*

- Constructed an event-driven CLI OS in Go with composition-root dependency injection and interface-driven storage, enabling a full infrastructure swap in 1 line while enforcing sync.RWMutex safety across 6+ concurrent modules.
- Built a RAG engine from scratch — OpenAI embeddings, hybrid vector search (SQLite-vec + custom cosine similarity), and GPT response synthesis across 4+ data sources — exposing 4 AI productivity commands via an async ingestion loop.

SKILLS

- **Languages:** Go (net/http, Gin, GORM), TypeScript, JavaScript (ES6+), Python
- **Backend & Cloud:** AWS (Lambda, Step Functions, DynamoDB, S3, EventBridge, SQS, CloudWatch), GCP (BigQuery), Node.js, Docker
- **Frontend:** React.js, Next.js, TypeScript, Tailwind CSS
- **Databases:** DynamoDB, PostgreSQL, MySQL, MongoDB, Redis, SQLite
- **AI & Tooling:** OpenAI API, Retrieval-Augmented Generation, Vector Search, Git, CI/CD (GitHub Actions)

ACHIEVEMENTS & OPEN SOURCE

- Active open-source contributor to Appwrite, Keyshade, and Discord Ticket — picking up open issues and engaging with maintainers on GitHub and Discord.
- Ranked in the Top 50 of 2,000+ candidates in HackerRank's Orchestrate hackathon.
- Certified in AWS Cloud Computing (Internshala) and Azure Functions (Microsoft Learn).